HW8. 540.

Prob. 1 (2&T, Ch. 6, 1.1)

Verify

(1) $$u\,\underline{\nabla}^2 w = \underline{\nabla}\cdot(u\,\underline{\nabla} w) - (\underline{\nabla} u)\cdot(\underline{\nabla} w)$$

Assume 3 spatial dim-ns

Sol-n.

Assuming 3 spatial dimensions, let

(2) $$\underline{\nabla} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

$w = w(x,y,z)$ & $u = u(x,y,z)$, then

u is distributed like a scalar as u is a scalar f-n

(3) $$\underline{\nabla} u = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle u = \left\langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right\rangle \quad \&$$

(4) $$\underline{\nabla}\cdot\underline{\nabla} u = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \left\langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right\rangle = u_{xx} + u_{yy} + u_{zz}$$

Then, working each side of (1) separately,

$$LHS = u\,\underline{\nabla}^2 w = u\cdot\underline{\nabla}\cdot\underline{\nabla} w = u\cdot(w_{xx} + w_{yy} + w_{zz}) =$$
$$= u\,w_{xx} + u\,w_{yy} + u\,w_{zz}.$$

$$RHS = \underline{\nabla}\cdot(u\cdot\underline{\nabla} w) - (\underline{\nabla} u)\cdot(\underline{\nabla} w) =$$
$$= \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \left\langle u\cdot\frac{\partial w}{\partial x}, u\cdot\frac{\partial w}{\partial y}, u\cdot\frac{\partial w}{\partial z} \right\rangle -$$
$$- \left\langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right\rangle \cdot \left\langle \frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}, \frac{\partial w}{\partial z} \right\rangle =$$
$$= \frac{\partial}{\partial x}\left[u\cdot\frac{\partial w}{\partial x}\right] + \frac{\partial}{\partial y}\left[u\cdot\frac{\partial w}{\partial y}\right] + \frac{\partial}{\partial z}\left[u\cdot\frac{\partial w}{\partial z}\right] -$$
$$- \left\{ \frac{\partial u}{\partial x}\cdot\frac{\partial w}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{\partial w}{\partial y} + \frac{\partial u}{\partial z}\cdot\frac{\partial w}{\partial z} \right\} =$$

$$\left[\underline{u_x w_x} + u w_{xx} + \underline{\underline{u_y w_y}} + u w_{yy} + \underset{\sim}{u_z w_z} + u w_{zz}\right] -$$

$$-\left\{\underline{u_x w_x} + \underline{\underline{u_y w_y}} + \underset{\sim}{u_z w_z}\right\} = u w_{xx} + u w_{yy} + u w_{zz} \equiv LHS \checkmark$$

hence verifies the given relation. $\blacksquare$

Prob.2. (2&T, Ch.6, 1.2 modif-d)

(?0) why do we need it?

Let u be in $C^2(\Omega)$ & and in $C^1(\Omega)$ where Ω is a "normal bounded domain" (means regular dom-n, i.e. Lipshitz, bounded, domain). Suppose that

(1) $$\nabla^2 u = 0 \qquad \Omega$$

(2) $$u = 0 \qquad S,$$

where $S = \partial\Omega$, the boundary of the dom-n. (i) Show that $u = 0$ in Ω. (ii) Explain what this means in terms of the steady state heat eq-n problem?

Hint: Assume $\underline{x} \in \mathbb{R}^3$ & note that this argument can easily be extended to $\mathbb{R}^n$. Use one of the Green's identities with $w = u$ & use the fact that if the integral of a cont-s nonnegative f-n over Ω is zero, the the f-n is identically zero in Ω.

Soln: [Preamble]

(i) By hint, the fact:

(3) $$\int_\Omega f(x)\, d\underline{x} = 0 \quad \text{for } f(\underline{x}) \geq 0, \ \& \ f(\underline{x}) \in C^0(\Omega) \Rightarrow$$

$$f(\underline{x}) \equiv 0.$$

Recall Green's first identity: outward flux surface integral, over tiny dS

(4) $$\int_\Omega u \cdot \nabla^2 w \, d\underline{x} = \int_{\partial\Omega} u \frac{\partial w}{\partial \underline{n}} dS - \int_\Omega \nabla u \cdot \nabla w \, d\underline{x}$$